
Who Knows the Answer They Want to Give? Pre-Commitment and Reactivity Across a Post-Training Gradient

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Abstract

A popular intuition holds that reinforcement-learning (RL) post-training makes large language models (LLMs) “lock in” the answer they intend to give, leaving them less able than base models to react to information that arrives mid-conversation. We test this hypothesis directly using a matched post-training gradient of one model family (QWEN2.5-1.5B: base $\rightarrow$ instruct/RLHF $\rightarrow$ RL-reasoning), with three experiments that measure behavioral reactivity to new information, the internal earliness of answer commitment via hidden-state probes, and the redistribution of answer uncertainty when a sub-fact is held certain. Our headline finding inverts the hypothesis: it is the *base* model that most “knows the answer it wants to give.” A linear probe decodes the base model’s own final answer from its prompt hidden state at 87% balanced accuracy (chance 25%), versus 71% for instruct and only 33% for the RL-reasoning model, which decides through generation rather than pre-encoding. Behaviorally, the base model is the most rigid, changing its answer just 2% of the time when handed a genuinely informative hint ($p \approx 10^{-27}$), while post-trained models revise $\sim 47\%$ of the time. Post-training thus *reduces* internal pre-commitment and *increases* reactivity, though plain RLHF over-corrects into sycophantic flipping. Holding a sub-fact certain redistributes uncertainty strongly for the base model (-0.32 bits, $p \approx 10^{-13}$) but not at all for the reasoning model. The “pre-plans, cannot react” deficit is real, but it belongs to base models and to information arriving after commitment.

1 Introduction

When a deployed language model is corrected, handed a fresh fact, or shown that its first answer was wrong, we want it to listen. Yet a growing intuition in the community is that the very training that makes models useful assistants—reinforcement learning (RL) from human feedback and, more recently, RL on verifiable rewards—also makes them stubborn: the model appears to decide where its turn is going before it starts writing, and then refuses to be moved. If true, this would be a direct reliability and safety problem. The models we put in front of users in high-stakes, multi-turn settings would be exactly the ones *least* able to absorb a correction once the conversation is underway—the “lost-in-conversation” failure.

The hypothesis. This paper tests a specific, three-part version of that intuition, posed as a research question: do RL/post-trained LLMs *pre-commit* to the answer they will give—encoding it early and resisting revision when new information arrives—more than their base counterparts? And can we *rotate* which sub-fact of a decomposable question is held certain to *redistribute* the model’s uncertainty onto the remaining parts? We separate this into three claims. **(C1)** Pre-planning exists: the final answer is decodable from internal state before it is emitted. **(C2)** RL increases pre-commitment and reduces reactivity, both internally (the answer is encoded earlier) and behaviorally (the model revises less). **(C3)** Rotating certainty redistributes uncertainty—and RL models redistribute less.

The gap. Each ingredient has been studied, but never together on the comparison that the hypothesis is actually about. Interpretability work firmly establishes C1: probes on the prompt hidden state predict a response’s length, answer, and correctness before the first token [Dong et al., 2025, Pal et al., 2023, Bigelow et al., 2025]. But that work does not isolate RL—it never contrasts a matched base model against its RL-trained descendant. Conversely, the behavioral literature on revision and sycophancy (e.g., “Try Again” prompting, self-correction, FlipFlop challenges) measures reactivity directly but never ties it to an internal pre-commitment probe, and it disagrees on the *direction* of the deficit: some report RL-induced *under*-reaction and answer repetition [Anonymous, 2025b, Kumar et al., 2024], others report widespread *over*-reaction that is not monotonic in RLHF [Laban et al., 2023]. Finally, C3 has no precedent: no prior work rotates what is held certain across a question to test whether uncertainty actually spreads.

Our approach. We supply the missing contrast. We assemble a matched post-training gradient of a *single* model family—QWEN2.5-1.5B (base), QWEN2.5-1.5B-INSTRUCT (SFT + RLHF/DPO), and DEEPSEEK-R1-DISTILL-QWEN-1.5B (distilled from R1’s RL reasoning traces)—so that architecture and tokenizer are held fixed and post-training intensity is the variable. On the *same* items we run three experiments: (1) a two-turn **reactivity** test that injects content-free pushback or a genuine hint and measures whether the model revises (C2b); (2) a **commitment-earliness probe** that decodes the model’s own final answer from its prompt hidden state and along the generation timeline (C1, C2a); and (3) a **rotating-certainty** protocol on decomposable STRATEGYQA questions that measures how much answer entropy moves when one sub-fact is held certain (C3).

What we find. The results invert the hypothesis. It is the *base* model, not the RL model, that most knows the answer it wants to give. The probe decodes the base model’s final answer from its prompt hidden state at **87%** balanced accuracy (chance 25%), falling to 71% for instruct and to near-chance **33%** for the RL-reasoning model. Behaviorally the base model is the most rigid: it revises only **2%** of the time when handed a genuinely informative hint, versus $\sim 47\%$ for both post-trained models ($p \approx 10^{-27}$). Holding a sub-fact certain redistributes the base model’s uncertainty by -0.32 bits ($p \approx 10^{-13}$) but leaves the reasoning model’s entropy essentially untouched ($p = 0.84$). Post-training does not increase pre-commitment—it moves the commitment point *later* and makes the model reactive, at the cost (for plain RLHF) of introducing the opposite failure, sycophantic over-reaction.

Contributions.

- We provide the first matched base-vs-RL *pre-planning-probe* contrast, decoding each model’s own answer from its prompt representation, and show that internal pre-commitment *decreases* monotonically with post-training ($0.87 \rightarrow 0.71 \rightarrow 0.33$), with the commitment point moving later along generation.
- We tie this internal measure to behavior on two tasks, showing that the base model is the consistently rigid one while the two post-trained models exhibit the two opposite reactivity failures the literature flagged—instruct over-reacts (sycophancy), the reasoning model is selective on ARC-CHALLENGE but over-reacts on GSM8K.
- We introduce and run the first test of *rotating certainty*, showing that holding a sub-fact certain redistributes answer uncertainty—strongly for the base model, marginally for instruct, and not at all for the reasoning model.
- We release code, raw outputs, and figures for all three experiments, with logged parse-failure rates and bootstrap/Wilcoxon statistics, so the contrast is reproducible.

The rest of the paper reviews related work (section 2), details our models, datasets, and metrics (section 3), presents the three experiments (section 4), and discusses what the inverted result means and where it stops (section 5).

2 Related Work

Our study sits at the intersection of four threads: internal pre-planning in language models, the effect of RL post-training on revision, behavioral sycophancy, and uncertainty decomposition. We organize related work by these themes and position our contribution against each.

Internal pre-planning (supports C1). A line of interpretability work shows that decoder-only LMs encode global properties of their *entire* upcoming response in early hidden states, despite next-token

training. Dong et al. [2025] train simple probes on the prompt hidden state that predict a response’s length, number of reasoning steps, final multiple-choice answer, and eventual correctness—before the first token—and report a “U-shaped” commitment trajectory along generation. Pal et al. [2023] establish the foundational primitive that future tokens are linearly decodable from a single hidden state, and Wu et al. [2024] formalize “pre-caching” versus “breadcrumbs” and show planning grows with scale. Anonymous [2025a] show that the reasoning-token budget of a reasoning model is linearly decodable and steerable, the most literal evidence for “planning the end of the turn”—but, like the rest of this thread, their subjects are all RL/distilled reasoning models with *no base contrast*. Bigelow et al. [2025] give a causal, circuit-level definition of planning and find that instruction tuning *selects among* existing plans rather than creating planning, and that base models can hold “competing plans.” We reuse the prompt-position probing protocol of Dong et al. [2025], but extend the target to the model’s *own* answer and, crucially, run it across a matched base-to-RL gradient—the comparison this thread omits.

RL post-training and revision (C2, contested). Anonymous [2025b] provide the strongest direct support for the under-reaction reading: single-turn RLVR collapses revision, with RL models repeating the identical answer across turns and the unique-answer ratio dropping after RL across methods and scales. Kumar et al. [2024] similarly find “behavior collapse,” where naive RL locks in the first answer, and introduce the clean $\Delta^{i \rightarrow c} / \Delta^{c \rightarrow i}$ decomposition we adopt to separate beneficial fixes from harmful flips. Yet the same work notes the base model can *over-react*, making the direction model-dependent. We connect this behavioral literature to an internal probe on matched models, and find that—measured internally—it is the base model, not the RL model, that pre-commits.

Sycophancy and change-of-mind (C2, behavioral). Laban et al. [2023] run challenger prompts (“are you sure?”) across ten models and report a 46% flip rate and -17% accuracy, with sycophancy *universal* across models trained with and without RLHF and *not* monotonic in RLHF—a direct warning against a naive “RL causes the deficit” reading. This over-reaction failure is the mirror image of pre-commitment; the two coexist and are selected by model and task. Our two-task design (ARC-CHALLENGE and GSM8K) reproduces both ends of this axis within one family.

Uncertainty decomposition (C3, open). Methods exist for measuring and decomposing LLM uncertainty into sources [Anonymous, 2025c] and for response-wise uncertainty over reasoning steps, but no prior work rotates which part of a decomposable question is held certain to test whether residual uncertainty redistributes. We introduce that protocol and report the first measurements of it.

Summary. Prior work establishes that planning exists (C1) and that revision can collapse *or* over-fire after training (C2), but never on a matched base-vs-RL probe contrast, and never tests rotating certainty (C3). We close all three gaps in one study.

3 Methodology

Our design isolates one variable—post-training intensity—and measures its effect on internal answer commitment, behavioral reactivity, and uncertainty redistribution on *identical* items.

3.1 Models: a matched post-training gradient

We use three checkpoints from one model lineage, sharing architecture and tokenizer so that the systematic difference between them is the amount and type of post-training (table 1). The BASE model is pretrained only; INSTRUCT adds supervised fine-tuning plus RLHF/DPO; and RL-REASONING is distilled from DeepSeek-R1’s RL reasoning traces. Because true base models are not served by any chat API, all weights are run locally. Each model is used in its *native* interface: BASE receives a plain transcript prompt (it has no chat template), while INSTRUCT and RL-REASONING use their own chat templates. We run bf16 weights on an NVIDIA RTX A6000, with greedy decoding for commitment measures and temperature 0.7 sampling for entropy. Seeds are fixed (42; 123 for Experiment 3 sampling).

3.2 Datasets

We use three pre-gathered local datasets chosen to fit each experiment. ARC-CHALLENGE (science multiple-choice) gives clean single-letter answers and is used for the reactivity and probe experi-

Table 1: The matched post-training gradient. All three models share the QWEN2.5-1.5B architecture and tokenizer lineage, so post-training stage is the only systematic difference.

Key	HuggingFace id	Post-training stage
BASE	Qwen/Qwen2.5-1.5B	none (pretrained only)
INSTRUCT	Qwen/Qwen2.5-1.5B-Instruct	SFT + RLHF/DPO
RL-REASONING	deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B	distilled from R1’s RL reasoning traces

ments. GSM8K (grade-school math) gives numeric answers and serves as a robustness check for reactivity. STRATEGYQA (decomposable yes/no questions with an accompanying facts chain) supports the rotating-certainty experiment, because each item can be split into two hops.

3.3 Experiment 1: behavioral reactivity (C2b)

We convert each single-turn item into a two-turn episode: the model answers, feedback is injected, and the model answers again. Feedback is one of three types. `generic` is content-free pushback (“that’s incorrect, try again”), following the “Try Again”/UFO protocol; `challenge` is the FlipFlop-style “are you sure?”; and `hint` is genuine information pointing toward the correct answer. We report the *answer change rate* from turn 1 to turn 2 and, to reveal *how* a model reacts, the directional flip rates $\Delta^{i \rightarrow c}$ (a wrong answer corrected to right—a beneficial fix) and $\Delta^{c \rightarrow i}$ (a right answer broken into wrong—harmful over-reaction). We run $N = 200$ ARC-CHALLENGE items per feedback type and $N = 150$ GSM8K items as a robustness check. Note that the GSM8K hint is non-informative (“re-check your arithmetic”), so on GSM8K all three feedback types probe reaction to *content-free* nudges; we flag this when interpreting.

3.4 Experiment 2: commitment-earliness probe (C1, C2a)

We greedily generate each model’s answer, parse the model’s *own* final letter, and train a balanced logistic-regression probe (80/20 split) to predict that letter from hidden states. We probe two locations: the **prompt-end** hidden state (across all layers), which measures how much of the answer is pre-encoded *before* generation, and a mid-layer trajectory *along* the generation timeline, which measures how early the answer becomes decodable. Probe accuracy at the prompt position far above the four-way chance level (0.25) indicates the answer is pre-encoded. Because the probe targets the model’s own answer rather than the gold answer, a model’s low task accuracy does not bias the measurement. We use $N = 440$ four-way (A–D) ARC-CHALLENGE items.

3.5 Experiment 3: rotating certainty (C3)

For each STRATEGYQA item we split the facts chain into two hops, A and B . We then compare three conditions: `baseline` (neither hop given), `+A` (hop A held certain in the prompt), and `+B` (hop B held certain). Our uncertainty measure is the predictive binary entropy of the yes/no answer, estimated from $K = 6$ samples per item, which is model-agnostic and does not depend on logit access. We report $H(\text{baseline})$, $H(+A)$, $H(+B)$, the signed reductions ΔH , and paired significance. A reduction in entropy when a hop is fixed indicates uncertainty was *redistributed* from that hop onto the rest of the question. We run $N = 120$ items.

3.6 Baselines and statistics

The within-study baseline for every comparison is the BASE model. Probe accuracy is compared against the random/majority chance level (0.25 for four-way ARC-CHALLENGE). We report bootstrap 95% confidence intervals (10k resamples), Fisher exact tests for the categorical change rates in Experiment 1, Wilcoxon signed-rank tests for the paired entropy reductions in Experiment 3, and Mann–Whitney U for cross-model comparisons. Parse-failure rates are logged ($\leq 6\%$, concentrated in the reasoning model) and reported rather than silently dropped.

Table 2: Experiment 1, ARC-CHALLENGE ($N = 200$). Turn-1→turn-2 answer **change rate** by feedback type, with bootstrap 95% CIs, and turn-1 accuracy. The base model is the most rigid on every feedback type and essentially ignores a genuine hint. Most rigid (lowest change) per column in **bold**.

Model	Turn-1 acc	generic	challenge	hint
BASE	0.65	0.17 [.12,.22]	0.43 [.35,.50]	0.02 [.01,.04]
INSTRUCT	0.72	0.52 [.44,.59]	0.56 [.49,.64]	0.47 [.40,.54]
RL-REASONING	0.37	0.47 [.41,.53]	0.43 [.36,.50]	0.46 [.39,.54]

Table 3: Experiment 1, ARC-CHALLENGE directional flip rates. $\Delta^{i \rightarrow c}$ is beneficial (wrong corrected to right); $\Delta^{c \rightarrow i}$ is harmful over-reaction (right broken to wrong). INSTRUCT over-reacts under content-free generic ($\Delta^{c \rightarrow i} = 0.33$, **bold**); RL-REASONING is selective, fixing more than it breaks under hint.

Model	generic		hint	
	$\Delta^{i \rightarrow c}$	$\Delta^{c \rightarrow i}$	$\Delta^{i \rightarrow c}$	$\Delta^{c \rightarrow i}$
BASE	0.06	0.04	0.02	0.00
INSTRUCT	0.11	0.33	0.25	0.20
RL-REASONING	0.24	0.11	0.30	0.08

4 Results

We present the three experiments in turn. Read together, the internal probe (section 4.2) explains the behavior seen in reactivity (section 4.1) and uncertainty (section 4.3): *the base model encodes its answer in the prompt, and everything else follows*.

4.1 Experiment 1: reactivity to new information

Table 2 reports the turn-1→turn-2 answer change rate on ARC-CHALLENGE. The base model barely revises, and it *almost entirely ignores a genuinely informative hint* appended after its answer: a 2% change rate under hint, versus 47% for both post-trained models. Fisher exact tests confirm the gap is enormous: BASE versus INSTRUCT and BASE versus RL-REASONING differ at $p = 2.6 \times 10^{-13}$ and 3.1×10^{-10} under generic, and at $p = 4.3 \times 10^{-28}$ and 3.8×10^{-27} under hint. Post-training does not make models more rigid; it makes them *far more* willing to revise.

How each model reacts. The change rate hides the *direction* of revision, which table 3 and figure 1 make explicit. INSTRUCT (plain RLHF) *over-reacts*: under content-free pushback it flips correct→incorrect 33% of the time (38% under challenge)—textbook sycophancy. The reasoning model is the most *selective* reactor: it fixes wrong answers far more than it breaks correct ones, especially given a hint ($\Delta^{i \rightarrow c} = 0.30$ versus $\Delta^{c \rightarrow i} = 0.08$). The base model does neither, because it barely moves at all.

Robustness on GSM8K. Table 4 repeats the experiment on numeric math. Turn-1 accuracy now favors the math-distilled reasoning model (RL-REASONING 0.77, INSTRUCT 0.67, BASE 0.22). Two findings emerge. First, the *base model is rigid on both tasks*—the single most consistent result across the paper. Second, the *direction* of post-training’s reactivity effect is task- and correctness-dependent: here the high-accuracy reasoning model *over-reacts*, re-deriving and breaking its mostly-correct solutions under content-free pushback ($\Delta^{c \rightarrow i}$ of 0.35–0.39), the mirror image of its selective behavior on ARC-CHALLENGE. Fisher tests put BASE/INSTRUCT versus RL-REASONING at $p \leq 5.5 \times 10^{-12}$ on every condition, while BASE versus INSTRUCT is not significant ($p > 0.3$). Because the GSM8K hint is non-informative, this task isolates reaction to content-free feedback.

4.2 Experiment 2: the answer is pre-encoded—in the base model

Table 5 reports the central mechanistic result. A probe predicting each model’s *own* final answer from its prompt hidden state reaches 0.873 balanced accuracy for the base model—more than three

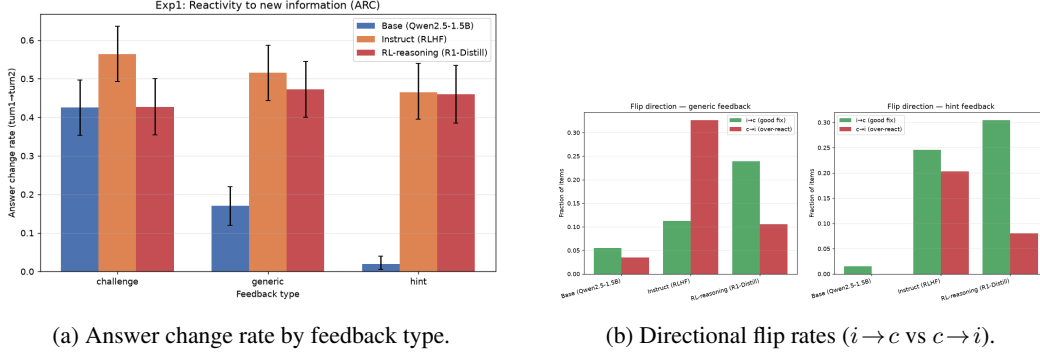


Figure 1: Experiment 1 on ARC-CHALLENGE. (Left) The base model barely revises and ignores a genuine hint, while both post-trained models revise about half the time. (Right) The *direction* differs: INSTRUCT breaks correct answers under content-free pushback (over-reaction), whereas RL-REASONING preferentially fixes wrong answers (selective reaction).

Table 4: Experiment 1 robustness on GSM8K ($N = 150$). All three feedback types are effectively content-free here. The base and instruct models are both rigid; the math-distilled RL-REASONING over-reacts, breaking correct solutions ($\Delta^{c \rightarrow i}$ 0.35–0.39). Lowest change per column in **bold**.

Model	Turn-1 acc	generic	challenge	hint *	Dominant flip
BASE	0.22	0.15	0.13	0.08	— (rigid)
INSTRUCT	0.67	0.16	0.18	0.07	— (rigid)
RL-REASONING	0.77	0.57	0.51	0.56	$c \rightarrow i$ 0.35–0.39

times chance—falling to 0.712 for instruct and to near-chance 0.331 for the reasoning model. The ordering is monotonic and inverts the hypothesis: *more post-training means the answer is committed later, not earlier*.

figure 2 shows where and when the answer becomes decodable. The base model’s answer emerges from the prompt representation in the late layers and is committed early in generation (0.72 decodable by 25% of the way through); instruct decides late (the curve rises only after 75%); and the reasoning model decides through generation, its answer reaching decodability only near the end. This is the clean, identically-measured contrast that the behavioral results reflect.

4.3 Experiment 3: rotating certainty redistributes uncertainty—in the base model

Table 6 shows that holding a sub-fact certain reduces answer entropy for the base model by 0.32 bits when hop *A* is fixed and 0.25 bits for hop *B*, both strongly significant ($p = 1.8 \times 10^{-13}$); uncertainty is redistributed from the fixed hop onto the rest of the question. The effect is marginal for instruct (~ 0.08 – 0.09 bits, $p = 0.019$) and *absent* for the reasoning model (≈ 0 bits, $p = 0.84$): injecting certainty about one hop does not move its answer entropy at all, because it reasons over the whole question regardless. The cross-model gap is large (BASE vs RL-REASONING reduction: Mann–Whitney $p = 8.6 \times 10^{-9}$).

5 Discussion

A single mechanism explains all three experiments. The probe (section 4.2) is the hinge. The base model encodes its answer in the prompt representation before it generates (0.87 decodable). This one fact predicts its behavior everywhere else. It is rigid: because the answer is already fixed by the prompt, feedback *appended after* the answer—whether empty pushback (0.17 change) or a genuine hint (0.02 change)—cannot move it (section 4.1). Yet its uncertainty is highly shaped by *what is in the prompt*: placing a fact *before* the answer redistributes its entropy by 0.32 bits (section 4.3). Both behaviors follow from the same cause—the answer is a function of the prompt representation. Information in the prompt shapes the commitment; information after it does not. This resolves

Table 5: Experiment 2, ARC-CHALLENGE ($N = 440$; four-way chance = 0.25). Probe balanced accuracy predicting the model’s *own* final answer from the prompt-end hidden state (best layer), and self-consistency (own answer vs gold). The base model pre-encodes its answer (**0.873**); the reasoning model does not (**0.331**, near chance). Most pre-encoded in **bold**.

Model	Prompt-position acc	Self-consistency (vs gold)
BASE	0.873	0.67
INSTRUCT	0.712	0.68
RL-REASONING	0.331	0.39

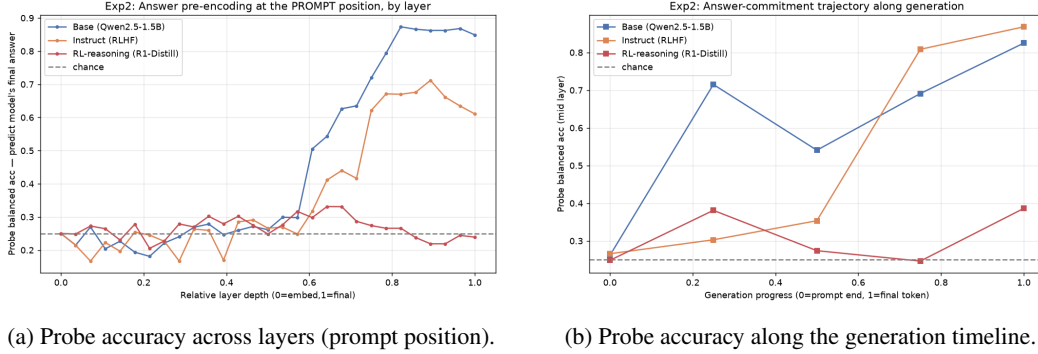


Figure 2: Experiment 2 on ARC-CHALLENGE. (Left) The base model’s answer becomes strongly decodable from the prompt hidden state in late layers (to 0.87); INSTRUCT reaches 0.71; the reasoning model stays near chance (0.25) at every layer—its answer is not pre-encoded. (Right) Commitment timing: the base model commits *early* (0.72 by 25% of generation), INSTRUCT commits *late* (rising only after 75%), and the reasoning model’s answer becomes decodable only at the very end. Commitment moves later with more post-training.

what looks like a contradiction between Experiment 1 (base ignores a hint) and Experiment 3 (base is strongly moved by a fact): the difference is whether the information arrives before or after the model has committed.

Post-training moves commitment later. Instruct decides during generation, which makes it reactive—but plain RLHF makes it *indiscriminately* so. It over-reacts, abandoning correct answers under content-free challenge ($\Delta^{c \rightarrow i}$ up to 0.38). This is the sycophancy/FlipFlop failure [Laban et al., 2023], the opposite of pre-commitment. The reasoning model commits latest of all (probe 0.33, at chance) and is both the most selective reactor and the most robust in its uncertainty: it reasons over the whole question rather than reading its answer off the prompt. On the hypothesis’s own terms, it “knows the answer it wants to give” the *least*.

Reconciling the contested direction. Our *internal* mechanism is clean and monotonic, but the *behavioral* reactivity direction among post-trained models depends on the task and on whether the model was right to begin with. On ARC-CHALLENGE, where models are often wrong and the hint is informative, the reasoning model reacts beneficially; on GSM8K, where it is usually right and feedback is content-free, the same model over-reacts and breaks correct solutions. What is invariant across tasks is *base-model rigidity*. This is exactly the “contested direction” the literature flagged [Anonymous, 2025b, Kumar et al., 2024, Laban et al., 2023]: under- and over-reaction co-exist and are selected by conditions. We therefore report both rather than claiming a single direction, and we locate the two classic failure modes at opposite ends of one post-training axis—UFO/SCoRe-style under-reaction in the *base* model, FlipFlop-style over-reaction in *instruct*.

Limitations. We are honest about the scope of these claims.

- **One family, one scale (1.5B).** Lineage is matched, but INSTRUCT and RL-REASONING differ in the *type* of post-training (RLHF vs reasoning distillation), not a clean RL ablation. The result should be confirmed at 7B and on other families (e.g., SimpleRL-Zoo, Qwen-UFO).

Table 6: Experiment 3 on STRATEGYQA ($N = 120$; entropy in bits). ΔH is the entropy *reduction* when a hop is held certain (larger = more redistribution). Fixing a hop strongly redistributes the base model’s uncertainty, marginally for INSTRUCT, and not at all for RL-REASONING. Largest redistribution per column in **bold**.

Model	$H(\text{base})$	$H(+A)$	$H(+B)$	$\Delta H(+A)$	$\Delta H(+B)$	Wilcoxon p
BASE	0.722	0.404	0.471	+0.318	+0.250	1.8×10^{-13}
INSTRUCT	0.420	0.341	0.327	+0.079	+0.093	0.019
RL-REASONING	0.555	0.607	0.528	−0.052	+0.027	0.84 (n.s.)

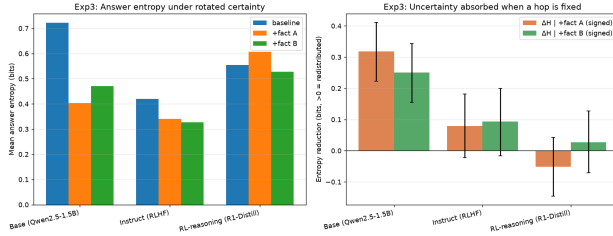


Figure 3: Experiment 3 on STRATEGYQA. Answer entropy under baseline, +A, and +B. Fixing a hop redistributes uncertainty strongly for the base model (large, significant drop with CIs clear of 0), marginally for INSTRUCT, and not at all for the reasoning model, whose answer entropy is robust to injected sub-fact certainty.

- **Interface differs by model** (base=transcript, others=chat), because base models have no chat mode. This is inherent to the comparison but a confound for the *absolute* reactivity rates in Experiment 1. The internal probe (Experiment 2), measured identically across models, is the cleaner contrast—and it agrees.
- **Low ARC-CHALLENGE accuracy for RL-REASONING (37%)** reflects a math-distilled model on science multiple-choice under a 640-token budget (reasoning is sometimes truncated). Because the probe targets the model’s own answer, this does not bias Experiment 2, but turn-1 accuracy should be read with care.
- **Coarse entropy.** Experiment 3 estimates entropy from $K = 6$ samples per item; we rely on 120-item aggregates and paired tests, which are robust, and replaced a noise-inflated per-item $|\Delta H|$ metric with signed bootstrap-CI reductions.
- **Heuristics.** Greedy decoding with a fixed token budget and sentence-level STRATEGYQA fact-splitting are heuristics; parse-failure rates ($\leq 6\%$, mostly the reasoning model) are logged and reported, never silently dropped.

Broader implications. The “pre-plans and cannot react” property that motivates safety concern is real, but it is a property of *base/pretrained* models and of information that arrives *after* the model has committed. RL post-training largely *mitigates* it—at the cost, for plain RLHF, of the opposite failure. Practically, this suggests the right lever for mid-conversation correctability is not avoiding RL but choosing a post-training recipe that decides late *and* discriminately, as the reasoning model does on ARC-CHALLENGE. It also suggests rotating certainty could become a decoding-time probe for residual per-hop uncertainty—useful precisely where a model’s answer is prompt-driven.

6 Conclusion

We set out to test whether RL post-training makes LLMs pre-commit to their answers and react less to new information, using a matched base→instruct→RL-reasoning gradient of a single model family. Within this family the hypothesis is **inverted**. RL/post-training does not increase answer pre-commitment or reduce reactivity—it *reduces* internal pre-commitment (the answer decoded from the prompt falls from 0.87 in the base model to 0.71 for instruct to 0.33, near chance, for the reasoning model) and *increases* reactivity. The “knows-the-answer / cannot-react” property is real, but it belongs to the *base* model—rigid on both ARC-CHALLENGE and GSM8K—and to informa-

tion that arrives *after* commitment. Among post-trained models the reactivity *direction* is task- and correctness-dependent (beneficially selective on ARC-CHALLENGE, over-reactive on GSM8K), so the deficit has no single direction; its only invariant is base-model rigidity. Finally, rotating certainty does redistribute uncertainty, confirming C3, but only where the answer is prompt-driven—strongly in the base model and not at all in the reasoning model.

Key takeaway. The model that most “knows the answer it wants to give” is the one with the *least* post-training, and the deficit is about *when* information arrives relative to the model’s internal commitment point.

Future work. Three directions follow. First, replicate at 7B and on a clean RL ablation (e.g., Qwen2.5-base vs SimpleRL-Zoo, vs Qwen-UFO) to separate RL from reasoning distillation. Second, causally steer the late-layer “answer direction” found in Experiment 2 to test whether forcing earlier commitment *induces* base-like rigidity. Third, extend rotating certainty to multi-hop QA (e.g., HotpotQA) and test whether it can serve as a decoding-time intervention that surfaces a reasoning model’s residual per-hop uncertainty.

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A Implementation Details

Hardware and decoding. All models are run locally in bf16 on a single NVIDIA RTX A6000. Commitment measures (Experiments 1 and 2) use greedy decoding; entropy (Experiment 3) uses temperature 0.7 sampling with $K = 6$ samples per item. Random seeds are fixed at 42 (123 for the Experiment 3 sampling). Generation budgets are capped at a fixed maximum token count per item (640 tokens for the reasoning model, which occasionally truncates its chain of thought on ARC-CHALLENGE).

Interfaces. The base model has no chat template and is prompted with a plain transcript format; the instruct and reasoning models use their respective chat templates and are run multi-turn. This difference is inherent to comparing a base model against post-trained descendants and is discussed as a confound for absolute reactivity rates in section 5; the probe of Experiment 2 is measured identically across all models.

Probe. The Experiment 2 probe is a balanced logistic regression trained with an 80/20 train/test split on last-token hidden states, evaluated at the prompt-end position across all layers and along a mid-layer generation trajectory. The target is the model’s own parsed final answer letter (A–D), so a model’s task accuracy does not bias the measurement.

Statistics. Confidence intervals are bootstrap 95% intervals over 10k resamples. Experiment 1 uses Fisher exact tests on the 2×2 change/no-change contingency tables; Experiment 3 uses Wilcoxon signed-rank tests for paired entropy reductions and Mann–Whitney U for cross-model comparisons.

B Data and Parse Failures

ARC-CHALLENGE (science multiple-choice, single-letter answers) is used for Experiments 1 and 2; GSM8K (numeric) is used for the Experiment 1 robustness check; STRATEGYQA (decomposable yes/no with a facts chain) is used for Experiment 3, with the fact chain split into two hops at the sentence level. Answer parsing uses a regex plus last-mention heuristic. Parse-failure rates are at most 6% and are concentrated in the reasoning model (whose long chains occasionally truncate before emitting a final letter). Failures are logged and reported, never silently dropped. Raw model outputs are released under `results/` and figures under `figures/`.